

Bone Stacking Combinatorial Number System

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Abstract

This paper defines a finite number system by enumerating combinatorial structures built from at most twelve abstract bones. A structure is first classified by its total number of bones, then by its structural partition, and finally by the orientations of its one-bone, two-bone, and three-bone piles. The resulting system contains exactly 2,911 distinct values. We derive the counting formula, give ranking and unranking procedures, and provide both concrete and generalized examples.

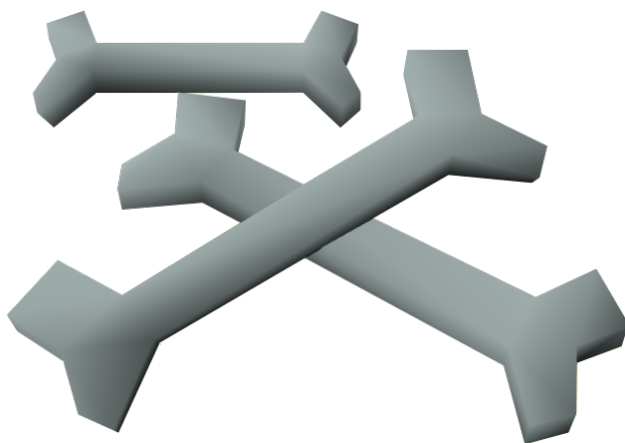


Figure 1: Bone number C1, representing decimal 9.

1 Introduction

The system was inspired by the observation that many small mammals have twelve pairs of ribs, as well as a curiosity surrounding the invention of roman numerals. The mathematical construction, however, treats the bones as abstract combinatorial objects. Up to twelve bones are used to represent a number, and are interpreted in a base-2911 number system to allow for individual digits to be read regardless of the positioning of the bones.

The central idea is to enumerate every distinct structure in a fixed order over three levels:

1. Structures are grouped by total number of bones: all one-bone structures, then all two-bone structures, and so on through twelve bones. This is to minimize the number of bones needed for small numbers, while expanding the total number of representable numbers.
2. Within a fixed bone-count, structures are grouped by their structural partition. For example, three bones can be partitioned as

$$1 + 1 + 1, \quad 2 + 1, \quad 3.$$

Because the piles are unordered, $1 + 2$ is not a separate partition from $2 + 1$.

3. Within a fixed partition, the possible orientations are enumerated according to a fixed canonical ordering.

This gives a one-to-one correspondence between valid bone structures and the natural numbers.

$$1, 2, \dots, 2911.$$

2 Definitions

Definition 2.1 (Pile). *A pile is a connected group of one, two, or three bones. Four bones is likely to topple over and is not considered. Bone piles can be represented by the alphabet: 1,A,B,C,D,E,F,G,H,I.*

Definition 2.2 (Bone orientation). *A one-bone pile has one orientation of bone stackings:*

$$1 = \text{No connections..}$$

A two-bone pile has three orientations:

$$\begin{aligned} A &= \text{end-to-end,} \\ B &= \text{end-to-middle,} \\ C &= \text{middle-to-middle.} \end{aligned}$$

A three-bone pile has six orientations:

$$\begin{aligned} D &= AA = \text{end-to-end} + \text{end-to-end,} \\ E &= AB = \text{end-to-end} + \text{end-to-middle,} \\ F &= AC = \text{end-to-end} + \text{middle-to-middle,} \\ G &= BB = \text{end-to-middle} + \text{end-to-middle,} \\ H &= BC = \text{end-to-middle} + \text{middle-to-middle,} \\ I &= CC = \text{middle-to-middle} + \text{middle-to-middle.} \end{aligned}$$

The ordering convention is

$$A < B < C, \quad D < E < F < G < H < I.$$

Definition 2.3 (Structural partition). *Suppose a structure consists of a one-bone piles, b two-bone piles, and c three-bone piles, where $a, b, c \in \mathbb{N}_0$ and $a + b + c \leq 12$. The structural partition of a structure containing n bones is the triple (a, b, c) satisfying*

$$a + 2b + 3c = n.$$

The piles are treated as unordered within each size class. Thus, for example, ABC represents the same structure regardless of the order in which the three two-bone piles are listed. The canonical representation is obtained by sorting orientations according to

$$A < B < C$$

for two-bone piles and

$$D < E < F < G < H < I$$

for three-bone piles.

When different pile sizes occur, three-bone piles are written first, followed by two-bone piles, followed by one-bone piles. Within a partition, smaller letters are written first.

Definition 2.4 (Bone-number). *A bone-number is the canonical symbolic representation of a valid structure. The decimal value of a bone-number is its position in the global enumeration described above.*

3 The Beginning of the Enumeration

The first few values are

$$\begin{array}{ll}
 1 = 1, & 9 = C1, \\
 2 = 11, & 10 = D, \\
 3 = A, & 11 = E, \\
 4 = B, & 12 = F, \\
 5 = C, & 13 = G, \\
 6 = 111, & 14 = H, \\
 7 = A1, & 15 = I, \\
 8 = B1, &
 \end{array}$$

Continuing,

$$\begin{array}{ll}
 16 = 1111, & 24 = BC1, \\
 17 = A11, & 25 = CC1, \\
 18 = B11, & 26 = D1, \\
 19 = C11, & 27 = E1, \\
 20 = AA1, & 28 = F1, \\
 21 = AB1, & 29 = G1, \\
 22 = AC1, & 30 = H1, \\
 23 = BB1, & 31 = I1.
 \end{array}$$

At the upper end of the system,

$$\begin{array}{l}
 2907 = HHHH, \\
 2908 = HHHI, \\
 2909 = HHII, \\
 2910 = HIII, \\
 2911 = IIII.
 \end{array}$$

Thus the largest value is

$$\boxed{2911},$$

4 Counting the Structures

The counting can be reduced to a simple multiset problem.

Lemma 4.1. *For a structural partition containing b two-bone piles, the number of possible orientation multisets is*

$$\binom{b+2}{2}.$$

Proof. There are three possible two-bone orientations, A, B, C . We must choose a multiset of size b from these three types. Equivalently, let $x_A, x_B, x_C \geq 0$ satisfy

$$x_A + x_B + x_C = b.$$

By stars and bars, the number of solutions is

$$\binom{b+3-1}{3-1} = \binom{b+2}{2}.$$

□

Lemma 4.2. *For a structural partition containing c three-bone piles, the number of possible orientation multisets is*

$$\binom{c+5}{5}.$$

Proof. There are six possible three-bone orientations,

$$D, E, F, G, H, I.$$

A multiset of size c is determined by nonnegative integers

$$x_D + x_E + x_F + x_G + x_H + x_I = c.$$

Again by stars and bars, the number of solutions is

$$\binom{c+6-1}{6-1} = \binom{c+5}{5}.$$

□

Proposition 4.3 (Structures in a fixed partition). *If a partition contains " a " one-bone piles, " b " two-bone piles, and " c " three-bone piles, then the number of distinct structures is*

$$\boxed{\binom{b+2}{2} \binom{c+5}{5}}.$$

Proof. The one-bone piles have only one possible orientation, since all one-bone piles are identical. The two-bone piles contribute $\binom{b+2}{2}$ orientations, and the three-bone piles contribute $\binom{c+5}{5}$ orientations. These choices are independent, so the total is their product. □

5 Counting an Entire Bone Block

Let B_n denote the number of structures containing exactly n bones.

For each pair (b, c) satisfying

$$2b + 3c \leq n,$$

where b is the number of 2-bone structures and c is the number of 3-bone structures, the remaining number of bones is

$$a = n - 2b - 3c.$$

We first fix the number c of 3-bone structures and consider every possible value of c , from 0 up to $\lfloor n/3 \rfloor$ since each multiple of 3-bone structures uses exactly 3 bones from the total n . For each fixed c , we then consider every possible number b of 2-bone structures that can be formed from the remaining bones, from 0 up to $\lfloor (n - 3c)/2 \rfloor$.

For each pair (b, c) , there are

$$\binom{b+2}{2}$$

ways to distribute the b 2-bone structures among their three possible types, and

$$\binom{c+5}{5}$$

ways to distribute the c 3-bone structures among their six possible types.

Therefore,

$$B_n = \sum_{c=0}^{\lfloor n/3 \rfloor} \sum_{b=0}^{\lfloor (n-3c)/2 \rfloor} \binom{b+2}{2} \binom{c+5}{5}.$$

Denote the cumulative number of structures through n bones.

$$C_n = \sum_{k=1}^n B_k$$

Theorem 5.1 (Finite enumeration). *The construction defines exactly 2911 distinct bone-numbers.*

Proof. Every valid structure is uniquely defined by a total bone count $n \in \{1, \dots, 12\}$, a structural partition (a, b, c) , and a canonical multiset of orientations within each pile size. The enumeration orders these disjoint categories without overlap. Hence each structure occurs exactly once.

The number of structures in the n -bone block is B_n , so the total number of structures is

$$\sum_{n=1}^{12} B_n = C_{12} = 2911.$$

□

6 Ranking and Unranking

The conversion between decimal integers and bone-numbers can be understood as a pair of inverse operations.

6.1 Decimal to Bone-Number

Let $N \in \{1, \dots, 2911\}$. First find the unique n satisfying

$$C_{n-1} < N \leq C_n,$$

where $C_0 = 0$. The number lies in the n -bone block. Its one-indexed position within that block is

$$r = N - C_{n-1}.$$

Next enumerate the structural partitions of n bones. For each partition, add

$$\binom{b+2}{2} \binom{c+5}{5}$$

until the partition containing position r is found. If the cumulative number before the selected partition is P , then its one-indexed position within that partition is

$$s = r - P.$$

Finally, unrank the orientation multisets within the selected partition. In a partition containing b two-bone piles and c three-bone piles, define

$$M_2(b) = \binom{b+2}{2}, \quad M_3(c) = \binom{c+5}{5}.$$

The partition therefore contains

$$M_3(c)M_2(b)$$

structures. Structures are ordered by first enumerating the c -multiset of three-bone orientations, then the b -multiset of two-bone orientations, then the trivial a -multiset one-bone orientation. The one-indexed ranks of these two components are

$$p = \left\lfloor \frac{s-1}{M_2(b)} \right\rfloor + 1$$

and

$$q = (s-1) \bmod M_2(b) + 1.$$

Thus p identifies the p -th three-bone orientation multiset, while q identifies the q -th two-bone orientation multiset.

6.2 Decimal to Bone-Number Example: 2500

Consider the decimal number

$$2500.$$

Since

$$1847 < 2500 \leq 2911,$$

the number lies in the twelve-bone block. Its position inside that block is

$$2500 - 1847 = 653.$$

The relevant partition table begins

Partition	Number of structures	Cumulative
1^{12}	1	1
$2 + 1^{10}$	3	4
$2^2 + 1^8$	6	10
$2^3 + 1^6$	10	20
$2^4 + 1^4$	15	35
$2^5 + 1^2$	21	56
2^6	28	84
$3 + 1^9$	6	90
$3 + 2 + 1^7$	18	108
$3 + 2^2 + 1^5$	36	144
$3 + 2^3 + 1^3$	60	204
$3 + 2^4 + 1$	90	294
$3^2 + 1^6$	21	315
$3^2 + 2 + 1^4$	63	378
$3^2 + 2^2 + 1^2$	126	504
$3^2 + 2^3$	210	714

Because

$$504 < 653 \leq 714,$$

the required partition is

$$3^2 + 2^3.$$

Its local rank is

$$653 - 504 = 149.$$

Each choice of the two three-bone orientations is followed by ten choices of the three two-bone orientations. Hence

$$g = \left\lfloor \frac{149 - 1}{10} \right\rfloor + 1 = 15$$

and

$$q = (149 - 1) \bmod 10 + 1 = 9.$$

The 15-th three-bone orientation is

$$FI,$$

while the 9-th two-bone orientation is

$$BCC.$$

Therefore

$$\boxed{2500 = FIBCC}.$$

6.3 Bone-Number to Decimal

The reverse process is a ranking operation. Given a bone-number, its corresponding decimal value is determined by finding its position in the complete enumeration of structures.

For a general bone-number, proceed as follows:

1. Count the total number of bones n .
2. Determine its structural partition (a, b, c) , where a , b , and c denote the numbers of one-, two-, and three-bone piles, respectively.
3. Add the complete blocks containing fewer than n bones. If C_{n-1} denotes the cumulative number of structures through $n - 1$ bones, this contributes C_{n-1} .
4. Within the n -bone block, enumerate the structural partitions in the prescribed order and add the sizes of all partitions preceding the selected partition. Let this cumulative number be P .
5. Determine the one-indexed rank s of the given orientation multiset within the selected partition.

If P is the number of structures preceding the selected partition and s is the one-indexed rank within that partition, then the corresponding decimal value is

$$\boxed{N = C_{n-1} + P + s.}$$

6.4 Bone-Number to Decimal Example: "IX"

See Figure 1 for an example illustration.

Let the observed bone placements I and X denote the one-, two-bone orientations respectively:

$$I = 1,$$

$$X = C.$$

Since bone placement permutations are considered equivalent, we use the canonical representation

$$IX = C1.$$

This has structural partition

$$2 + 1,$$

Which contains three bones. The structures with fewer than three bones are

$$11, A, B, C.$$

Thus the cumulative number of structures through two bones is

$$C_2 = 5.$$

Within the three-bone block, the structural partitions preceding $2 + 1$ consist only of

$$1 + 1 + 1.$$

This partition contains exactly one structure:

$$111.$$

Therefore,

$$P = 1.$$

Within the $2 + 1$ partition, the possible two-bone orientations are

$$A1, B1, C1.$$

Thus $C1$ is the third structure in this partition, so

$$s = 3.$$

Therefore the global rank is

$$N = C_2 + P + s = 5 + 1 + 3 = 9.$$

Hence

$$\boxed{IX'' = C1 = 9}.$$

7 A Flappy Bird Example

7.1 Encoding Game Stat and Program

The system can also be used as an abstract representation for the state of a simple game such as *Flappy Bird*. For example, one could assign bone-numbers to quantities such as

1. X -coordinate,
2. Y -coordinate,
3. velocity,
4. gravity,
5. flap impulse,
6. pipe X -positions,
7. pipe gap Y -positions,
8. score,
9. game speed,
10. random-number state.

Each state variable could be encoded using one or more values in the range $1, \dots, 2911$.

A deliberately conservative estimate of roughly 190 instructions and 10 states would require

$$200 \log_2(2911) \approx 2301$$

bits, or

$$\frac{2301}{8} \approx 288 \text{ bytes.}$$

This estimate represents the amount of memory needed to store game states and program instructions while excluding graphics, input/output, and other machine states. Since small mammals are imagined to provide two racks of ribs, this would correspond to 100 small mammals.

7.2 Encoding a 20×20 Monochrome Image

A 20×20 monochrome image contains

$$20 \cdot 20 = 400$$

binary pixels, and therefore has

$$2^{400}$$

possible states.

Since one bone-number has 2911 possible values, the minimum number m of bone-numbers required satisfies

$$2911^m \geq 2^{400}.$$

Taking logarithms gives

$$m \geq \frac{400}{\log_2(2911)} \approx 34.76.$$

Hence

$$\boxed{m = 35}.$$

So 35 bone-numbers are sufficient to encode every possible 20×20 monochrome image. Since each small mammal is imagined to provide two racks of ribs, this would correspond to 18 small mammals

8 Conclusion

The proposed number system is a finite combinatorial enumeration. A value is identified by the hierarchy

$$\boxed{\text{total bone count} \longrightarrow \text{structural partition} \longrightarrow \text{orientation multiset.}}$$

The number of structures for a fixed partition follows from multisets and stars and bars, giving

$$\binom{b+2}{2} \binom{c+5}{5}.$$

Summing over all partitions with at most twelve bones produces exactly

$$\boxed{2911}$$

distinct values.

The system therefore provides a well-defined bijection between the natural numbers $1, \dots, 2911$ and a finite family of canonical bone structures. Decimal integers can be *unranked* into bone-numbers by locating their block, partition, and orientation rank; bone-numbers can be *ranked* back into integers by reversing the same sequence.

9 Take Away

Given a set of 12 ribs from a small mammal, you could arrange the bones in piles of 1, 2, and 3, dispersed over a small region, such that you represent the numbers from 1 to 2911. This alphabet of 2911 numerical values is made of only 10 primitive symbols, limited structurally, with equivalent permutations.